Eunseong Bae

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Research Interests

- Manifold learning, Operator spectral theory, Asymptotic statistics

Academic Positions

- Postdoctoral Researcher, Research Institute of Applied Statistics, Sungkyunkwan University, South Korea, 2026–Present

Education

- Ph.D., Statistics, University of California, Davis, CA, USA, 2026 (Advisor: Wolfgang Polonik)
- M.S., Statistics, Seoul National University, South Korea, 2019 (Advisor: Chae Young Lim)
- B.S., Mathematical Sciences, Seoul National University, South Korea, 2014

Publications

- Working papers:
  - **Bae, E.**, Polonik, W., 2026. First-order asymptotic expansions for spectral convergence of normalized graph Laplacians constructed from samples on manifolds. Manuscript under final review.
- Preprints:
  - **Bae, E.**, Polonik, W., 2026. First-order asymptotic expansions for spectral convergence of compact self-adjoint operators on general spectral subsets, with application to kernel Gram matrices. arXiv: 2602.00999.
  - **Bae, E.**, Polonik, W., 2026. Kernel smoothing on manifolds. arXiv: 2601.16777.
- Published:
  - Kim, J., **Bae, E.**, Kim, Y., Lim, C.Y., Hur, J.-W., Kwon, J.S., Lee, S.-H., 2022. A robust multivariate structure of interindividual covariation between psychosocial characteristics and arousal responses to visual narratives. PLOS ONE 17, e0263817. DOI: 10.1371/journal.pone.0263817.
  - **Bae, E.**, Hur, J.-W., Kim, J., Kwon, J.S., Lee, J., Lee, S.-H., Lim, C.Y., 2020. Multi-group analysis using generalized additive kernel canonical correlation analysis. Sci Rep 10, 12624. DOI: 10.1038/s41598-020-69575-x.
- Theses:
  - **Bae, E.**, 2026. First-order Asymptotic Theory for Spectral Convergence of Graph Laplacians on Manifolds. Ph.D. Dissertation, University of California, Davis. Link: escholarship.org/uc/item/3nd8w4qd.
  - **Bae, E.**, 2019. Multi-group Analysis using Generalized Additive Kernel Canonical Correlation Analysis. Master's thesis, Seoul National University. Link: s-space.snu.ac.kr/handle/10371/161659.

Teaching Experience

- Teaching Assistant, University of California, Davis, 2019-2026
  - Graduate: Mathematical Statistics, Applied Statistics
  - Undergraduate: Elementary Statistics, Mathematical Statistics, Applied Statistics
- Teaching Assistant, Seoul National University, 2017
  - Undergraduate: Elementary Statistics

Work Experience

- Associate, Actuarial and Risk Management Team, Samsung Life Insurance, South Korea, 2014-2016

Professional Service

- Ad Hoc Reviewer, IEEE Transactions on Information Theory, 2026

Awards

- Teaching Recognition Award, Department of Statistics, UC Davis (2021; 2023)